

CS6175: VIRTUAL REALITY AND GAME-ENGINE TECHNOLOGIES

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Virtual Reality and Game-Engine Technologies

Subject Code

CS - Computer Science

Course Number

6175

Academic Unit

Computer Science (CS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

Virtual reality emphasizes on the construction of interactive 3D virtual environments, and how to interact with such environments through different sensory channels, such as audio, vision and gesture. Virtual Reality has many applications. The most popular one is 3D computer gaming, which is attracting a lot of attention in recent years. This course aims at introducing advanced virtual reality techniques and their applications. In particular, it investigates the 3D computer gaming application and the game engine architectures.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if DEC-A1 app.)		DEC-A2	DEC-A3
1	Identify the important characteristics of different virtual reality techniques.		x		
2	Evaluate different types of virtual reality systems.			x	
3	Explain and discuss different game engine architectures.			x	
4	Design and apply virtual reality techniques to address real-world problems.			x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lecture	Student will engage in formal lectures to gain knowledge about virtual reality technologies and their applications, including a very popular application of virtual reality - game engines.	1, 2, 3, 4	2 hours/ week
2	Tutorials	Students will participate in different class exercises that are relevant to virtual reality technologies and applications. Some of the tutorial exercises will involve the evaluation and design of virtual reality technologies.	1, 2, 3	1 hour/ week

3	Project	Students will participate in groups to consolidate their learning as they produce a program/report, integrating virtual reality techniques in an application under a game engine framework.	2, 3, 4	3 hours/week for 7 weeks
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Assessment Tasks / Activities (ATs)

ATs		CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Quiz	1, 2, 3	20	-	No
2	Course Project	2, 3, 4	20	-	Yes

Continuous Assessment (%)

40

Examination (%)

60

Examination Duration (Hours)

2

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

For a student to pass the course, at least 30% of the maximum mark for the examination must be obtained.

Assessment Rubrics (AR)**Assessment Task**

Quiz (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capacity in understanding the key concerns of virtual reality and game-engine techniques

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Note even reaching marginal levels

Assessment Task

Course Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to apply virtual reality and game-engine techniques to develop an application

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Note even reaching marginal levels

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to evaluate virtual reality and game software and to apply virtual reality and game-engine techniques on applications

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Note even reaching marginal levels

Assessment Task

Quiz (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Capacity in understanding the key concerns of virtual reality and game-engine techniques

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate

Failure

(F) Note even reaching marginal levels

Assessment Task

Course Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to apply virtual reality and game-engine techniques to develop an application

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate

Failure

(F) Note even reaching marginal levels

Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to evaluate virtual reality and game software and to apply virtual reality and game-engine techniques on applications

Excellent

(A+, A, A-) High

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Failure

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Part III Other Information

Keyword Syllabus

Real-time Rendering Techniques. Animation Techniques. Obstacle Avoidance. Emotion and Expressions. Physically Based Modeling. Motion Capture. Tracking Techniques. Display Systems. Game Engine Architecture. Virtual Reality Applications. Graphics Systems. Audio Rendering.

Syllabus

- Virtual Reality Technologies Overview of input and output devices for VR: head-mounted display, data gloves, 3D video capture, 3D displays, CAVE, haptic devices, motion tracking.;
- Interaction Techniques in Virtual Reality 3D selection and manipulation techniques, 3D user interface design and evaluation, gesture recognition and tangible interfaces.;
- Virtual Environments and Distributed Virtual Environments Real-time rendering techniques, visibility determination, motion prediction, motion synchronization.;
- Applications of Virtual Reality Applications of VR in different areas such as training, simulation and information visualization.;
- Game Engine Technologies Game Engine Architecture, Single- and Multi-Player techniques.

Reading List

Compulsory Readings

Title	
1	Nil

Additional Readings

Title	
1	G. Burdea and P. Coiffet, “Virtual Reality Technology,” Second Edition, Wiley-Interscience, 2003.
2	Mel Slater, Anthony Steed, and Yiorgos Chrysanthou, “Computer Graphics and Virtual Environments,” Addison Wesley, 2002.
3	Jobe Maker, “ActionScript for Multiplayer Games and Virtual Worlds,” New Riders, 2010.
4	Jason Busby, Zak Parrish, and Jeff Wilson, “Mastering Unreal Technology, Volume I: Introduction to Level Design with Unreal Engine 3,” Sams Publishing, 2010.
5	Jason Busby, Zak Parrish, and Jeff Wilson, “Mastering Unreal Technology, Volume II: Advanced Level Design Concepts with Unreal Engine 3,” Sams Publishing, 2010.